CURIO CLASS

Grade 4 math, unit 11.1: area and perimeter formulas

Five lesson days (89 to 93) with print, keys and checks

Draft for a teacher's read, November 2026

How to read this

This is one week of grade 4 math: five days of about 60 minutes each, on the standard 4.MD.A.3, applying the area and perimeter formulas for rectangles. Students come in counting squares and edges from grade 3 and leave writing an equation for a rectangle, solving for a missing side, working with bigger and fraction sides, and choosing which measure a situation needs.

WHAT IS IN THE PACK

- **The unit opener**, the math for the adult, with the common errors (A to F) the keys refer to.
- **The readiness check** (end of day 88) and its key.
- **For each day, 89 to 92:** the teacher page, the student page (printed both sides, at its real size), the key and the exit slip.
- **Day 91's generator sheet note**, which explains the practice sheet printed from the site.
- **Day 93, the end check:** teacher page, the check itself and its key.
- **The record sheet** for the whole unit and **the family note**.

Left out to keep the pack short: the projector slides for each day and the large-print student pages.

WHAT IS NOT READY YET

- Day 90 uses an interactive tool, *Community Plot Planner*, which is not built yet. Its teacher page says so at the top. The day's paper path (grid paper and the student page) is in the pack and can be taught as it stands; the tool will come later.
- The family note has a Spanish version. It waits for a qualified reader before it goes home.
- Nothing here has been taught yet. Every page is a draft.

FIVE QUESTIONS FOR YOU

1. Could you teach day 1 tomorrow from these pages alone? If not, what is missing?
2. Which day holds more than its minutes? Which holds less?
3. Where would your class get stuck, and does the plan say what to do there?
4. Is anything on the checks unfair, unclear, or not what the days taught?
5. What would you cut, and what would you never cut?

Notes in the margins are welcome. Short answers are fine.

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Unit 11.1: Area and perimeter formulas (days 89 to 93)

The math, for the adult

Standard 4.MD.A.3: "Apply the area and perimeter formulas for rectangles in real world and mathematical problems." Students met both measures in grade 3 by counting: area as the unit squares that cover a shape, arranged in rows and columns (3.MD.C.7), and perimeter as the length of the edge (3.MD.D.8). This unit turns each count into a formula and then uses the formula backwards.

Area is rows times columns. A rectangle 6 feet by 8 feet holds 6 rows of 8 one-foot squares, so its area is $6 \times 8 = 48$ square feet. $A = l \times w$ says the same thing for every rectangle. The unit is a square (a square foot, a square meter), because what is counted is squares.

Perimeter is the four sides added. $P = l + w + l + w$. Because there are two of each side, the same sum is $2 \times l + 2 \times w$, and by the distributive property $2 \times l + 2 \times w = 2 \times (l + w)$. In words: length plus width is half the perimeter. Students should see this on a drawn rectangle (the edge cut into two equal halves, each one length and one width) before they see it in symbols. The unit is a length (feet, meters), because what is measured is a line.

A missing side is a missing number in an equation. If a floor covers 63 square feet and is 7 feet wide, the equation is $7 \times ? = 63$, so the missing side is a missing factor: 9 feet. This is the unknown-factor work of 3.OA.A.4 and the factor pairs of 4.OA.B.4. If a fence is 38 feet all the way around and the pen is 12 feet long, the equation is $12 + ? + 12 + ? = 38$: the two long sides use 24 feet, the 14 feet left make two short sides, so each is 7 feet. Students write the equation for this rectangle, with its numbers and one unknown, and solve it. They do not recall a formula and fill it in. That is what "apply" means here (Progressions for the Common Core State Standards, Geometric Measurement).

Bigger and harder numbers. Day 91 uses sides of two and three digits (find $l + w$, then add the sum to itself) and sides that are fractions with like denominators ($1\ 5/6 + 2\ 2/6 + 1\ 5/6 + 2\ 2/6$). Day 92 asks students to decide which measure a situation needs, with no cue word. A cue word is any word that names the inside or the edge (covers, fills, around, edge, side, border of, whole floor); the situation (fence, tape, paint, trim, wire) is not a cue.

Limits. Rectangles only. Areas stay within 100, or a one-digit number times a multiple of 10. A side found from a perimeter comes from an even perimeter of 100 or less. Larger areas come back in stretches 12 to 14.

Why this model

Grid paper and tiles on day 89 make both counts visible: the squares inside, a finger along the edge. On day 90 the squares go away. Community Plot Planner gives each student a lot and five family notes ("we need 48 square feet; our plot is 6 feet wide"), so the student has to write the equation before the plot can be built; the squares fade as the student works. Where the tool is not built or there are no devices, the same notes are done on paper with grid paper and sketches. From day 91 a rectangle is a sketch with labelled sides, which is what students will meet on every later page.

The errors to expect, in plain words

	What students do	What it shows	What to do when many do it
A	Use the area formula for perimeter, or the other way round	The words have no pictures yet	Sort ten situations by drawing what is measured: the inside or the edge
B	Add one length and one width and call it the perimeter	They do not see the edge going all the way round	Trace and name all four sides; write l + w + l + w
C	Find a side from an area by subtracting, adding or multiplying the two numbers given	They do not see a missing factor	Rows of the known side on grid paper until the area is used up
D	From a perimeter, take off one side only, or forget to halve	They count one pair of sides	Fill in 14 + ___ + 14 + ___ = 44 together
E	Write "40 feet" for an area	Squares and lengths are not told apart	Eight answers with no units: write feet or square feet on each
F	Count the squares along the edge, not the unit sides (12 for a 5 by 3 rectangle)	The edge is not yet seen as unit lengths	Tick and number each unit side along the edge, then count the ticks

Eight answers with no units (error E): the fence round a yard, 64; the carpet on a bedroom floor, 120; the ribbon round a card, 30; the paint on a wall, 96; the tape round a rug, 26; the grass in a park, 900; the rope round a play area, 80; the tiles on a hall floor, 54. Students write feet or square feet (yards for the park) on each and say why.

Each day's teacher page names the wrong answer that signals each error on that day's exit.

The reteach (readiness gaps)

The readiness check is the last ten minutes of day 88. If more than a third of the class missed item 1 (area of a 6 by 7 rectangle) or item 2 (perimeter of a 9 by 4 rectangle), day 90's warm-up becomes this reteach, and the facts check moves to day 91.

1. Each student draws a 5 by 3 rectangle on grid paper.
2. Count the squares inside together, row by row: "5, 10, 15. Three rows of 5 is 15 squares." Write 3 × 5 = 15.
3. Trace the edge with a finger, naming each side: "5, 3, 5, 3." Write 5 + 3 + 5 + 3 = 16.
4. Ask: "Which number tells how much paper the rectangle covers? Which tells how far an ant walks around it?"
5. Students do a 4 by 6 rectangle alone, then compare with a partner.

For item 3 (8 × ? = 48) gaps: fact help goes into the warm-ups only, not into the lesson.

Supplies for the unit

- Per student: grid paper (about six sheets over the unit; large squares for students who need them), about 15 tiles or paper squares (day 89), a pencil.

- Per class: the projector for the display pages, a board, one string loop 36 units long (day 92's pens).
- Devices for day 90 if Community Plot Planner is built; otherwise grid paper.
- A room with nothing: hand-ruled grids and squares cut from paper do every job.

Words

Area, perimeter, length, width, square unit (square foot, square meter), formula, equation. Teach each on a labelled rectangle. "Covers" and "goes around" help at first; by day 92 problems use neither.

How the unit is checked

Readiness check: day 88, four items, marked before day 89. Exits: one each on days 89 to 92, written on the record sheet by error letter. End check: day 93, seven items, 30 minutes, unaided, no grid; items 5 and 6 are judged by the teacher with look-fors and answered in day 94's first five minutes. Area returns with larger sides in stretches 12 to 14; without that return the standard is only half taught.

Credit

Ours. Informed by, not adapted from: Illustrative Mathematics K-5, grade 4, unit 5 (CC BY 4.0, https://curriculum.illustrativemathematics.org/k5/), and the Progressions for the Common Core State Standards, Geometric Measurement.

Curio Class, grade 4 math, unit 11.1, opener, version draft 1, not yet taught, CC BY 4.0, curioclass.org

Ready for area and perimeter

Name _____ Date _____

Show your work for each one.

1. A rectangle is 6 units by 7 units. Find its area. Write the multiplication.

Answer: _____

2. A rectangle is 9 cm long and 4 cm wide. Find its perimeter.

Answer: _____

3. 8 × ? = 48

Answer: _____

4. On grid paper, draw two rectangles with perimeter 12 and different areas.

Draw on your grid paper. Write the area inside each.

Curio Class, grade 4 math, unit 11.1, readiness check, version draft 1, not yet taught, CC BY 4.0, curioclass.org

Ready for area and perimeter: key

The last ten minutes of day 88, alone, on paper, with grid paper for item 4. Read each item aloud once. Mark before day 89 and write each result on the record sheet (R1 to R4). If more than a third of the class missed item 1 or item 2, day 90's warm-up becomes the reteach in the unit opener. If fewer than a third missed items 1 and 2, day 89 builds one rectangle, not three. Item 3 gaps go to the warm-ups only.

#	Part	Question	Answer	Wrong answers and what they mean
1	3.MD.C.7a	A rectangle is 6 units by 7 units. Find its area. Write the multiplication.	6 × 7 = 42	26: error A, uses one formula for the other measure
2	3.MD.D.8	A rectangle is 9 cm long and 4 cm wide. Find its perimeter.	26 cm	13: error B, adds one length and one width; 36: error A, uses one formula for the other measure
3	3.OA.A.4	8 × ? = 48	6	40 or 56: (not seen as a missing factor); 384: (not seen as a missing factor)
4	3.MD.D.8	On grid paper, draw two rectangles with perimeter 12 and different areas.	Two of: 1 by 5, 2 by 4, 3 by 3	

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Day 89: Both measures of one rectangle

Teacher page

Day	89, unit 11.1, understanding day
Goal	I can find both measures of one rectangle and write each as a formula
Standard parts	4.MD.A.3 (a) the area formula, a side unknown (on a branch or Longer only); 4.MD.A.3 (f) the perimeter formula as 2 × (l + w): length plus width is half the perimeter; 4.MD.A.3 (g) the area formula, both sides known
Tool	None
Materials	per student, 15 tiles or paper squares cut from grid paper, grid paper (2 sheets), stretch 10's marked end check; per class, the projector with the display page, 15 large tiles for the teacher, and a board (no document camera: draw the grid on the board)
Print	student page, 1 per student, both sides; exit slip, 1 per student; facts check part 1 sheet from stretch 1, 1 per student; the key, 1 for the teacher
Leans on	courses/math-4/u11-1/opener.md; courses/math-4/u01-1/facts-check-1.md; courses/math-4/u11-1/record.md; docs/courses/math-4/s10.md
Version	draft 1, not yet taught
Credit	ours

Before class

Print the student page both sides, the facts check part 1 sheet, the exit slips and the key. Count 15 tiles or paper squares into a cup for each student. Have stretch 10's end checks ready to hand back, items 7 and 8 marked. Put the display page on the projector.

The day

5 min. Warm-up: facts check, part 1

Alone. The 36 products from stretch 1, untimed. Collect at 5 minutes; an item not reached is left blank, not marked wrong. Mark the sheets after class. Students under 32 of 36 keep their 9 by 9 table out from day 90.

5 min. Day 88's items 7 and 8

Hand back day 88's end checks. Show a "meets" answer for each on the board. Item 7: "b = 20 + 4 × 20. 4 × 20 = 80. 20 + 80 = 100 beads." Item 8: "7 × 7 is about 50, and 18 is about 20, so about 70. 175 is far too big. s = 67." Students fix their own in pen beside the first try. Collect them for the record sheet. If many gave 80 on 7 or took 175 on 8, each warm-up problem from day 90 starts with an estimate, for a week.

20 min. Teaching first: one rectangle

12 minutes together: build one rectangle (Build a rectangle 5 tiles long and 3 tiles wide. Find its area and its perimeter. **[5 × 3 = 15 square units; 5 + 3 + 5 + 3 = 16 units, or 2 × (5 + 3) = 16]**). Each

student builds along with tiles on grid paper and fills the frame on side 1 as you go, not ahead of you. In the box on side 1, they sketch the 5 by 3 and label its sides. The display shows the figure, then each step.

1. Build it: a row of 5 tiles, then 2 more rows of 5 under it. Each tile is one square unit.
2. Area is how many squares cover the inside. Count by fives: 5, 10, 15.
3. That is 3 rows of 5. Write it with the sides: A = l × w, so 5 × 3 = 15 square units.
4. Perimeter is the length all the way around the edge. Run a finger along it and count the unit sides: 16.
5. Write it side by side: P = l + w + l + w, so 5 + 3 + 5 + 3 = 16 units.
6. Stop halfway around: one length and one width, 5 + 3 = 8. The other half is the same 8.
7. So P = 2 × (l + w): 2 × (5 + 3) = 2 × 8 = 16 units. The same answer.

Say as you point: area **covers** the inside, perimeter **goes around** the edge. Leave A = l × w, P = l + w + l + w and P = 2 × (l + w) on the board.

For you: 2 × l + 2 × w = 2 × (l + w) is the distributive property (unit opener); students show it on the tiles, not by name.

Watch for a student who counts the squares along the edge instead of the unit sides (12, not 16: a corner square has two sides on the edge). Have them mark each unit side with a tick and count the ticks.

8 minutes in pairs. "Make any other rectangle from your 15 tiles. Each of you writes A = and both forms of P; swap and check that both forms give the same number."

15 min. On paper: three rectangles

"Three rectangles", side 1 (Draw a rectangle 6 units long and 2 units wide. Write both equations. **[6 × 2 = 12 square units; 6 + 2 + 6 + 2 = 16 units, or 2 × (6 + 2) = 16]**, Draw a rectangle 9 units long and 5 units wide. Write both equations. **[9 × 5 = 45 square units; 9 + 5 + 9 + 5 = 28 units, or 2 × (9 + 5) = 28]**, Draw a rectangle 10 units long and 3 units wide. Write both equations. **[10 × 3 = 30 square units; 10 + 3 + 10 + 3 = 26 units, or 2 × (10 + 3) = 26]**). Alone. Students draw each one on grid paper, shade the inside, trace the edge, then write both equations with the units. For each perimeter they write the four sides first, then the 2 × form. Tiles stay on the desk only to check the 6 by 2.

Watch for a perimeter that is one length and one width (8, 14 or 13: error B), answers in the wrong rows (error A) and edge squares counted (24 or 22: error F). Ask: "Did you count the squares or the edge? Which one did the question ask for?" If many give one side pair: trace all four sides together and write l + w + l + w. If many swap them: sort the three drawings' numbers into "covers" and "goes around" on the board.

If the readiness check showed no gap (fewer than a third missed item 1 and fewer than a third missed item 2), this part is one rectangle (Draw a rectangle 6 units long and 2 units wide. Write both equations. **[6 × 2 = 12 square units; 6 + 2 + 6 + 2 = 16 units, or 2 × (6 + 2) = 16]**), 5 minutes. Then "Missing side" on side 2 (A tray covers 42 square inches. It is 6 inches wide. How long is it? **[6 × ? = 42; 7 inches]**), alone, checked by multiplying back, 5 minutes. Then pairs start the planned question below on their own rectangle, 5 minutes; the class finishes it in "Pull it together".

5 min. On paper: fix two

"Fix it", side 2 (Ava says: "A rug is 6 feet long and 5 feet wide. Its area is 22 square feet." **[6 × 5 = 30 square feet. Ava added the four sides: 22 feet is the perimeter]**, Ben says: "A card is 10 inches long

and 7 inches wide. Its perimeter is 17 inches." **[10 + 7 + 10 + 7 = 34 inches. Ben added one length and one width]**). Pairs for one minute on Ava's, then the class. The display shows each wrong working beside the right one. Ava added the four sides: that is the edge, not the inside. Ben stopped halfway around.

5 min. Pull it together

Whole class.

Ask. "Is 2 × (l + w) always the same as l + w + l + w? How do you know?" Expect, and show in this order: "It worked for every pair's" (true for those; ask "and one we have not drawn?"); "you go one length and one width, then the same again" (trace the tile rectangle to the halfway corner and stop); "it is two lengths and two widths either way". Close on the second: halfway around is l + w, and the other half matches.

5 min. Exit: the slip

Alone, the exit slip (A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each. **[8 × 3 = 24 square meters; 8 + 3 + 8 + 3 = 22 meters (2 × (8 + 3) is also right)]**). Students write both equations and the units. Record on the record sheet under E89.

Other rooms

- **No tiles.** Students shade 15 squares on grid paper instead; the build is the same.
- **One projector.** You build at the front; the class builds first, then checks against yours.
- **One child at home.** Measure a real table or rug to the nearest foot and write both measures, then do side 1.

Shorter or longer

Shorter: drop the third rectangle and the pairs minute on Ava's. Never drop the four-side equation before the 2 × form, or the fixes. Longer: "Missing side" (A tray covers 42 square inches. It is 6 inches wide. How long is it? **[6 × ? = 42; 7 inches]**), then the challenge (Draw two rectangles with an area of 20 square units. Give them different perimeters. Find each perimeter. **[Two of: 1 by 20, perimeter 42; 2 by 10, perimeter 24; 4 by 5, perimeter 18]**).

Words and supports

Word Wall cards: area, perimeter, length, width, square unit, formula, equation, each on a labelled rectangle. Frames on the display: "The area is __ square units because __ rows of __ cover it." "The perimeter is __ units because __ + __ + __ + __ = __." Access: large-square grid paper; tiles for touch.

Tomorrow

On the slip, 11 (error B): in day 90's Example 2, have these students trace the pen's four sides on the display before you write 12 + ? + 12 + ? = 38. Area and perimeter swapped, 22 and 24 (error A): these students sit near you for day 90's examples; if many, day 90's warm-up goes to Ava's rug again and the facts check moves to day 91. 18 for the perimeter (error F): the same seats. "24 meters" (error E): note it; day 91 teaches it.

Credit

Ours. Informed by, not adapted from: Illustrative Mathematics K-5, grade 4, unit 5, lesson 16 (CC BY 4.0, https://curriculum.illustrativemathematics.org/k5/), and the Progressions for the Common Core State Standards, Geometric Measurement.

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Day 89: Both measures of one rectangle

Name _____ Date _____

I can find both measures of one rectangle and write each as a formula.

Build it. Build a rectangle 5 tiles long and 3 tiles wide. Find its area and its perimeter.

Area: count the squares. A = l × w: ____ × ____ = ____ square units

Perimeter: count the unit sides around the edge. P = l + w + l + w: ____ + ____ + ____ + ____ = ____ units

The same perimeter: P = 2 × (l + w): 2 × (____ + ____) = ____ units

Three rectangles. Draw each one on grid paper. Write both equations with units.

1. Draw a rectangle 6 units long and 2 units wide. Write both equations.

   A = ____ × ____ = ____ square units

   P = ____ + ____ + ____ + ____ = ____ units; 2 × (____ + ____) = ____

2. Draw a rectangle 9 units long and 5 units wide. Write both equations.

   A = _____

   P = _____

3. Draw a rectangle 10 units long and 3 units wide. Write both equations.

   A = _____

   P = _____

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Fix it. Each answer is wrong. Say what went wrong. Then do it right.

1. Ava says: "A rug is 6 feet long and 5 feet wide. Its area is 22 square feet."

   What went wrong: _____

2. Ben says: "A card is 10 inches long and 7 inches wide. Its perimeter is 17 inches."

   What went wrong: _____

Missing side. Write the equation, solve it, then check it.

A tray covers 42 square inches. It is 6 inches wide. How long is it?

Equation: _____ Answer: _____ Check: _____

Challenge. Draw two rectangles with an area of 20 square units. Give them different perimeters. Find each perimeter.

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Day 89: Both measures of one rectangle: key

Each answer is worked out in full. The last column names the wrong answers to expect and the error each one shows.

Where	Id	Part	Question	Answer	Wrong answers and what they mean
example	s11-K89-1	4.MD.A.3 (g), MP.8	Build a rectangle 5 tiles long and 3 tiles wide. Find its area and its perimeter.	5 × 3 = 15 square units; 5 + 3 + 5 + 3 = 16 units, or 2 × (5 + 3) = 16	
fix	s11-F89-1	4.MD.A.3 (g), MP.8	Ava says: "A rug is 6 feet long and 5 feet wide. Its area is 22 square feet."	6 × 5 = 30 square feet. Ava added the four sides: 22 feet is the perimeter	22: error A, uses one formula for the other measure (the wrong answer shown)
fix	s11-F89-2	4.MD.A.3 (f), MP.8	Ben says: "A card is 10 inches long and 7 inches wide. Its perimeter is 17 inches."	10 + 7 + 10 + 7 = 34 inches. Ben added one length and one width	17: error B, adds one length and one width (the wrong answer shown)
practice	s11-P89-1	4.MD.A.3 (g), MP.8	Draw a rectangle 6 units long and 2 units wide. Write both equations.	6 × 2 = 12 square units; 6 + 2 + 6 + 2 = 16 units, or 2 × (6 + 2) = 16	8: error B, adds one length and one width (one length and one width); area 16, perimeter 12: error A, uses one formula for the other measure (swapped)
practice	s11-P89-2	4.MD.A.3 (g), MP.8	Draw a rectangle 9 units long and 5 units wide. Write both equations.	9 × 5 = 45 square units; 9 + 5 + 9 + 5 = 28 units, or 2 × (9 + 5) = 28	14: error B, adds one length and one width (one length and one width); 24: error F, counts the squares along the edge, not the unit sides (squares along the edge)
practice	s11-P89-3	4.MD.A.3 (g), MP.8	Draw a rectangle 10 units long and 3 units wide. Write both equations.	10 × 3 = 30 square units; 10 + 3 + 10 + 3 = 26 units, or 2 × (10 + 3) = 26	13: error B, adds one length and one width (one length and one width); 22: error F, counts the squares along the edge, not the unit sides (squares along the edge)
practice	s11-P89-4	4.MD.A.3 (a), MP.2	A tray covers 42 square inches. It is 6	6 × ? = 42; 7 inches	36 or 48: error C, finds a side from an area by subtracting,

			inches wide. How long is it?		adding, or multiplying the two numbers given; 252: error C
challenge	s11-C89-1	4.MD.A.3 (g), MP.8	Draw two rectangles with an area of 20 square units. Give them different perimeters. Find each perimeter.	Two of: 1 by 20, perimeter 42; 2 by 10, perimeter 24; 4 by 5, perimeter 18	
exit (paper)	s11-X89	4.MD.A.3	A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each.	8 × 3 = 24 square meters; 8 + 3 + 8 + 3 = 22 meters (2 × (8 + 3) is also right)	11: error B, adds one length and one width; swapped: error A, uses one formula for the other measure; units swapped: error E, wrong unit: "40 feet" for an area; 18: error F, counts the squares along the edge, not the unit sides (squares along the edge)

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Name _____ Day 89

1. A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each.

_____

_____

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Name _____ Day 89

1. A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each.

_____

_____

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Name _____ Day 89

1. A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each.

_____

_____

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Name _____ Day 89

1. A rectangle is 8 meters long and 3 meters wide. Find the area and the perimeter. Write the equation and the units for each.

_____

_____

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Day 90: Find the missing side

Teacher page

Not ready to teach: tools/community-plot-planner/community-plot-planner.html missing.

Day	90, unit 11.1, understanding day
Goal	I can find a missing side from the area or the perimeter
Standard parts	4.MD.A.3 (a) the area formula, a side unknown; 4.MD.A.3 (b) the perimeter formula, a side unknown; 4.MD.A.3 (f) the perimeter formula as 2 × (l + w): length plus width is half the perimeter (on a branch or Longer only); 4.MD.A.3 (g) the area formula, both sides known (on a branch or Longer only)
Tool	Community Plot Planner (key), 25 min
Materials	per student, grid paper (1 sheet) and a device with Community Plot Planner open in Lot planner; per class, the projector with the display page, and a board (no document camera: draw it on the board)
Print	student page, 1 per student, both sides; exit slip, 1 per student (paper path only); facts check part 2 sheet from stretch 1, 1 per student; the key, 1 for the teacher
Leans on	tools/community-plot-planner/community-plot-planner.html; courses/math-4/u11-1/opener.md; courses/math-4/u01-1/facts-check-2.md; courses/math-4/u11-1/record.md
Version	draft 2, not ready, not yet taught
Credit	ours

Before class

Print the student page both sides, the facts check part 2 sheet and the key. Open Community Plot Planner on each device in Lot planner. Put the display page on the projector.

The day

5 min. Warm-up: facts check, part 2

Alone. The 36 quotients from stretch 1, untimed. Collect at 5 minutes; an item not reached is left blank, not marked wrong. Mark the sheets after class. After the facts check, students under 32 of 36 on day 89 keep their 9 by 9 table out for the lesson.

If the readiness check showed gaps (more than a third missed item 1 or 2): the warm-up goes to the reteach in the unit opener, with five minutes from "On your own" and the partner check; the facts check moves to day 91, and those two questions are dropped; day 91's fixes 3 and 4 ask the same two parts.

If day 89's slips showed many A (area and perimeter swapped): Ava's rug again, "covers" or "goes around"; the facts check moves to day 91. With the reteach also here, the reteach comes first and A goes to day 91's warm-up.

10 min. Teaching first: two worked examples

Whole class. Students fill the two frames on side 1 as you go, not ahead of you. The display shows each figure, then each step on its own screen.

Example 1 (A garden bed covers 63 square feet. It is 7 feet wide. How long is it? **[7 × ? = 63; 9 feet]**). The missing side is a missing factor. Each square stands for one square foot.

1. Sketch the bed: 7 feet wide, the length unknown, 63 square feet inside.
2. Area is how many squares cover it. Each square stands for one square foot, and they come in rows of 7.
3. How many rows of 7 make 63? Write 7 × ? = 63.
4. Count by sevens: 7, 14, 21, 28, 35, 42, 49, 56, 63. That is 9 rows.
5. 7 × 9 = 63, so the bed is 9 feet long.
6. Check: 7 × 9 = 63.

Example 2 (A goat pen has 38 feet of fence all the way around. It is 12 feet long. How wide is it? **[12 + ? + 12 + ? = 38; 7 feet]**).

1. Sketch the pen: 12 feet on both long sides, because opposite sides of a rectangle match. The width is unknown on both short sides.
2. The fence goes all the way round: 12 + ? + 12 + ? = 38.
3. The two long sides use 12 + 12 = 24 feet.
4. 38 − 24 = 14 feet are left for the two short sides together.
5. Each short side is half of 14: 7 feet.
6. Check: 12 + 7 + 12 + 7 = 38.
7. Another way: length plus width is half the fence. Half of 38 is 19, and 19 − 12 = 7. The same answer.

Write the four-side equation first, every time: it stops errors B and D.

20 min. Community Plot Planner: the five notes

Alone, in Lot planner. Each family note gives one number and one side; the student finds the missing side, writes the equation, then tags the plot with the measure the note did not give. Students write each fence equation as four sides in the Plot notes table on side 2, even where the tool shows 2 × (5 + ?) = 30 (The Lee family needs 48 square feet. Their plot is 6 feet wide. How long is it? **[6 × ? = 48; 8 feet; fence 28 feet]**, The Nguyen family needs 36 square feet. Their plot is 4 feet wide. How long is it? **[4 × ? = 36; 9 feet; fence 26 feet]**, The Clark family needs 60 square feet. Their plot is 5 feet wide. How long is it? **[5 × ? = 60; 12 feet; fence 34 feet]**, The Ortiz family puts 30 feet of fence all the way around their plot. The plot is 5 feet wide. How long is it? **[5 + ? + 5 + ? = 30; 10 feet; area 50 square feet]**, The Abebe family puts 24 feet of fence all the way around their plot. The plot is 8 feet long. How wide is it? **[8 + ? + 8 + ? = 24; 4 feet; area 32 square feet]**).

Watch for a tag marked "counted" or a plot "built with squares" (the pieces came back), and a missing side that is the note's two numbers added or taken away (54 or 42 for the Lee family: error C). Ask as you walk: "Which number is fixed? Which side is missing? What equation says that?"

Stop at 20 minutes, lot finished or not: the notes, equations and tags are the job.

15 min. On paper: fix two, then two alone

"Fix it", side 1 (Jo says: "A rug covers 35 square feet. It is 5 feet wide, so it is 30 feet long." **[5 × ? = 35; 7 feet. Jo subtracted]**, Max says: "A goat pen has 44 feet of fence all the way around. It is 14 feet long, so it is 30 feet wide." **[14 + ? + 14 + ? = 44; 8 feet. Max took off one side only]**), 6 minutes. For Jo's, one minute of turn-and-talk in pairs, then the class. The display shows each wrong working beside the right one. Jo subtracted: area is rows times columns, so ask "5 times what is 35?". Max took off one long side and stopped: a fence has two long sides and two short ones.

"On your own", side 2 (A table top covers 27 square feet. It is 3 feet wide. How long is it? **[3 × ? = 27; 9 feet]**, A dog yard has 42 feet of fence all the way around. It is 13 feet long. How wide is it? **[13 + ? + 13 + ? = 42; 8 feet]**), 6 minutes, alone, no grid. Then 3 minutes in pairs: partners swap and check each answer by putting it back in the equation ("Does 3 × 9 give 27?"). A mismatch gets a question mark, not the answer.

If many miss the area question: rows of 3 on grid paper until 27 squares are used. If many miss the fence question: fill in 13 + __ + 13 + __ = 42 together.

5 min. Pull it together

Whole class.

Ask. "For the garden bed we never took anything away. For the goat pen we took away and then halved. Why?" Expect: "area is rows times columns, so it is what times 7"; "the fence has two of each side". Show the first, then the second; both began with an equation for this rectangle with one unknown.

Ask. "How can you check a missing side with no one to tell you?" Expect: put it back in the equation.

5 min. Exit: On your own in Community Plot Planner, or the slip

Alone. With devices: "On your own" from the tool's header, no hints, the same two parts as the slip with other numbers (A plot is 56 square feet and 7 feet wide. How long is it? **[8 feet]**, A plot is 18 square feet and 3 feet wide. How long is it? **[6 feet]**, A plot has 40 feet of fence all the way around. It is 12 feet long. How wide is it? **[8 feet]**, A plot has 22 feet of fence all the way around. It is 7 feet long. How wide is it? **[4 feet]**). Students show the end screen; write area and perimeter results on the record sheet under E90. No devices: the exit slip (A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation. **[3 × ? = 21; 7 inches]**, A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation. **[17 + ? + 17 + ? = 46; 6 inches]**).

Other rooms

- **No devices, or the tool not yet built.** The 20 minutes become the paper path: the five notes in the table on side 2, area notes shaded in rows on grid paper, fences sketched with four labelled sides, each checked by multiplying back or adding the four sides. Watch for errors C and D; ask the same three questions.
- **One projector.** You run the tool; the class writes each equation on side 2 before you type it.
- **One child at home.** Build the first "On your own" rectangle from 27 paper squares in rows of 3, then write the equation.

Shorter or longer

Shorter: drop the partner check (students check their own) and the challenge. Never drop the fixes or the written equations, and never give the grid back for "On your own". Longer: the challenge (A rectangle has an area of 40 square feet and a perimeter of 26 feet. How long are its sides? **[5 feet and 8 feet]**), then the extension (Find every rectangle with whole-number sides and an area of 36 square feet. Find the perimeter of each. **[1 by 36, perimeter 74; 2 by 18, 40; 3 by 12, 30; 4 by 9, 26; 6 by 6, 24]**).

Words and supports

Word Wall cards: area, perimeter, length, width, square foot, equation. Frames on the display: "The missing side is __ because __ × __ = __." "The two short sides are __ feet together, so each one is __ feet."

Tomorrow

On the slip, 18 or 24, or 63, on the card (error C): day 91 takes fix 3 first, whole class; on the end screen the same sign is 49 or 63, or 15 or 21. On the slip, 29 or 12 on the frame (error D): day 91 takes fix 4 first; on the end screen 28 or 16, or 15 or 8. Students with a tag marked "counted", or who shaded every square one by one on paper, sit near you for day 91's first ten minutes.

Credit

Ours. Informed by, not adapted from: Illustrative Mathematics K-5, grade 4, unit 5, lessons 16 and 17 (CC BY 4.0, https://curriculum.illustrativemathematics.org/k5/), and the Progressions for the Common Core State Standards, Geometric Measurement.

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Day 90: Find the missing side

Name _____ Date _____

I can find a missing side from the area or the perimeter.

Example 1. A garden bed covers 63 square feet. It is 7 feet wide. How long is it?

Equation: ___ × ___ = ___ The bed is ___ feet long. Check: ___ × ___ = ___

Example 2. A goat pen has 38 feet of fence all the way around. It is 12 feet long. How wide is it?

Equation: ___ + ___ + ___ + ___ = ___ The pen is ___ feet wide.

Check: _____

Fix it. Each answer is wrong. Say what went wrong. Then do it right.

1. Jo says: "A rug covers 35 square feet. It is 5 feet wide, so it is 30 feet long."

   What went wrong: _____

2. Max says: "A goat pen has 44 feet of fence all the way around. It is 14 feet long, so it is 30 feet wide."

   What went wrong: _____

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Plot notes. Write each equation. For a fence, write the four sides.

Note	Equation	Missing side	The other measure
Lee: 48 square feet, 6 feet wide			fence:
Nguyen: 36 square feet, 4 feet wide			fence:
Clark: 60 square feet, 5 feet wide			fence:
Ortiz: 30 feet of fence, 5 feet wide			area:
Abebe: 24 feet of fence, 8 feet long			area:

On your own. Write the equation, solve it, then check it.

1. A table top covers 27 square feet. It is 3 feet wide. How long is it?

   Equation: _____ Answer: _____ Check: _____

2. A dog yard has 42 feet of fence all the way around. It is 13 feet long. How wide is it?

   Equation: _____ Answer: _____ Check: _____

Challenge. A rectangle has an area of 40 square feet and a perimeter of 26 feet. How long are its sides? Sides: _____ and _____

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Day 90: Find the missing side: key

Each answer is worked out in full. The last column names the wrong answers to expect and the error each one shows.

Where	Id	Part	Question	Answer	Wrong answers and what they mean
example	s11-K90-1	4.MD.A.3 (a), MP.2	A garden bed covers 63 square feet. It is 7 feet wide. How long is it?	7 × ? = 63; 9 feet	
example	s11-K90-2	4.MD.A.3 (b), MP.2	A goat pen has 38 feet of fence all the way around. It is 12 feet long. How wide is it?	12 + ? + 12 + ? = 38; 7 feet	
tool	s11-T90-1	4.MD.A.3 (a), MP.2	The Lee family needs 48 square feet. Their plot is 6 feet wide. How long is it?	6 × ? = 48; 8 feet; fence 28 feet	54 or 42: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given
tool	s11-T90-2	4.MD.A.3 (a), MP.2	The Nguyen family needs 36 square feet. Their plot is 4 feet wide. How long is it?	4 × ? = 36; 9 feet; fence 26 feet	
tool	s11-T90-3	4.MD.A.3 (a), MP.2	The Clark family needs 60 square feet. Their plot is 5 feet wide. How long is it?	5 × ? = 60; 12 feet; fence 34 feet	
tool	s11-T90-4	4.MD.A.3 (b), MP.2	The Ortiz family puts 30 feet of fence all the way around their plot. The plot is 5 feet wide. How long is it?	5 + ? + 5 + ? = 30; 10 feet; area 50 square feet	
tool	s11-T90-5	4.MD.A.3 (b), MP.2	The Abebe family puts 24 feet of fence all the way around their plot. The plot is 8 feet long. How wide is it?	8 + ? + 8 + ? = 24; 4 feet; area 32 square feet	
fix	s11-F90-1	4.MD.A.3 (a), MP.2	Jo says: "A rug covers 35 square feet. It is 5 feet wide, so it is 30 feet long."	5 × ? = 35; 7 feet. Jo subtracted	30: error C, finds a side from an area by subtracting, adding, or multiplying the two

					numbers given (the wrong answer shown)
fix	s11-F90-2	4.MD.A.3 (b), MP.2	Max says: "A goat pen has 44 feet of fence all the way around. It is 14 feet long, so it is 30 feet wide."	14 + ? + 14 + ? = 44; 8 feet. Max took off one side only	30: error D, from a perimeter, takes off one side, or does not halve (the wrong answer shown)
practice	s11-P90-1	4.MD.A.3 (a), MP.2	A table top covers 27 square feet. It is 3 feet wide. How long is it?	3 × ? = 27; 9 feet	24 or 30: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given; 81: error C
practice	s11-P90-2	4.MD.A.3 (b), MP.2	A dog yard has 42 feet of fence all the way around. It is 13 feet long. How wide is it?	13 + ? + 13 + ? = 42; 8 feet	29: error D, from a perimeter, takes off one side, or does not halve (one long side taken off); 16: error D (not halved)
challenge	s11-C90-1	4.MD.A.3 (f), MP.2	A rectangle has an area of 40 square feet and a perimeter of 26 feet. How long are its sides?	5 feet and 8 feet	
challenge	s11-C90-2	4.MD.A.3 (g), MP.2	Find every rectangle with whole-number sides and an area of 36 square feet. Find the perimeter of each.	1 by 36, perimeter 74; 2 by 18, 40; 3 by 12, 30; 4 by 9, 26; 6 by 6, 24	
exit (paper)	s11-X90a	4.MD.A.3 (a)	A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation.	3 × ? = 21; 7 inches	18 or 24: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given; 63: error C
exit (paper)	s11-X90b	4.MD.A.3 (b)	A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation.	17 + ? + 17 + ? = 46; 6 inches	29 or 12: error D, from a perimeter, takes off one side, or does not halve
exit (tool)	s11-X90t1	4.MD.A.3 (a), MP.2	A plot is 56 square feet and 7 feet wide. How long is it?	8 feet	49 or 63: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given

exit (tool)	s11-X90t2	4.MD.A.3 (a), MP.2	A plot is 18 square feet and 3 feet wide. How long is it?	6 feet	15 or 21: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given
exit (tool)	s11-X90t3	4.MD.A.3 (b), MP.2	A plot has 40 feet of fence all the way around. It is 12 feet long. How wide is it?	8 feet	28: error D, from a perimeter, takes off one side, or does not halve (one long side taken off); 16: error D (not halved)
exit (tool)	s11-X90t4	4.MD.A.3 (b), MP.2	A plot has 22 feet of fence all the way around. It is 7 feet long. How wide is it?	4 feet	15: error D, from a perimeter, takes off one side, or does not halve (one long side taken off); 8: error D (not halved)

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Name _____ Day 90

1. A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation.

_____

_____

2. A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation.

_____

_____

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Name _____ Day 90

1. A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation.

_____

_____

2. A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation.

_____

_____

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Name _____ Day 90

1. A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation.

_____

_____

2. A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation.

_____

_____

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Name _____ Day 90

1. A card covers 21 square inches. It is 3 inches wide. How long is it? Write the equation.

_____

_____

2. A frame has 46 inches of edge all the way around. It is 17 inches long. How wide is it? Write the equation.

_____

_____

Curio Class, grade 4 math, day 90, version draft 2, not ready, not yet taught, CC BY 4.0, curioclass.org

Day 91: Both formulas, big sides and fraction sides

Teacher page

Day	91, unit 11.1, fluency day
Goal	I can use both formulas with no picture, with big sides and fraction sides
Standard parts	4.MD.A.3 (a) the area formula, a side unknown; 4.MD.A.3 (b) the perimeter formula, a side unknown; 4.MD.A.3 (c) perimeter with multi-digit sides; 4.MD.A.3 (d) perimeter with fraction sides; 4.MD.A.3 (e) choosing the measure in a real problem; 4.MD.A.3 (f) the perimeter formula as 2 × (l + w): length plus width is half the perimeter (on a branch or Longer only); 4.MD.A.3 (g) the area formula, both sides known
Tool	None
Materials	per student, a pencil and the day's generator sheet (no grid paper from today); per class, the projector with the display page, and a board (no document camera: draw it on the board)
Print	student page, 1 per student, both sides; generator sheet (sheet-d091), 1 per student, and its key, 1 for the teacher; exit slip, 1 per student; the key, 1 for the teacher; facts check part 2 sheet from stretch 1, 1 per student, only if day 90 ran the reteach
Leans on	courses/math-4/u11-1/opener.md; courses/math-4/u11-1/sheet-d091.md; courses/math-4/u01-1/facts-check-2.md; courses/math-4/u11-1/record.md
Version	draft 1, not yet taught
Credit	ours

Before class

Print what the header lists (the facts check part 2 only if day 90 ran the reteach). Put the display page on the projector. Put the grid paper away: from today a rectangle is a sketch with labelled sides.

The day

5 min. Warm-up: four sums

Alone, top of side 1: two multi-digit sums and two fraction sums (386 + 386 = ? **[772]**, 419 + 275 = ? **[694]**, 2 3/5 + 2 3/5 = ? **[5 1/5; 4 6/5 is also right]**, 3 5/8 + 1 7/8 = ? **[5 4/8; 4 12/8 and 5 1/2 are also right]**). Read the answers aloud at 4 minutes. Denominators added (4 6/10 for the third) is a fraction slip. Those students, and those day 90 flagged ("counted", or shading square by square), sit near you for the examples.

If day 90 ran the reteach: the facts check part 2 is the warm-up instead, untimed, collected at 5 minutes. The four sums are dropped; the examples carry the same moves.

If many made error C or D on day 90's exit: in "Fix it", take fix 3 (C) or fix 4 (D) first, whole class, with one minute's turn-and-talk. If day 90 sent A here (day 89's slip), Ava's rug is the warm-up. The facts check, when it runs, has the whole warm-up.

10 min. Teaching first: two worked examples

Whole class. Students fill the two frames on side 1 as you go. No grid: each sketch is a rectangle with all four sides labelled. The display shows each figure, then each step on its own screen.

Example 1 (A school field is 264 feet long and 158 feet wide. How much fence goes all the way around? **[264 + 158 = 422; 422 + 422 = 844; 844 feet]**).

1. Sketch the field: a rectangle, 264 feet on both long sides and 158 feet on both short sides. No grid, just labels.
2. The fence goes all the way around: 264 + 158 + 264 + 158.
3. Length plus width first: 264 + 158 = 422. That is half the fence.
4. The other half is the same, so add the sum to itself: 422 + 422 = 844.
5. The fence is 844 feet. It goes around, so the unit is feet, not square feet.
6. Check: 264 + 264 = 528 and 158 + 158 = 316. 528 + 316 = 844.

Example 2 (A picture frame is 6 3/4 inches long and 2 3/4 inches wide. Find its perimeter. **[6 3/4 + 2 3/4 = 8 6/4; 8 6/4 + 8 6/4 = 16 12/4 = 19; 19 inches (16 12/4, 18 4/4 and 76/4 are also right)]**).

1. Sketch the frame: 6 3/4 inches on both long sides, 2 3/4 inches on both short sides.
2. Length plus width: add the wholes, 6 + 2 = 8, and the fourths, 3/4 + 3/4 = 6/4. That is 8 6/4.
3. Add the sum to itself: 8 6/4 + 8 6/4. Wholes: 8 + 8 = 16. Fourths: 6/4 + 6/4 = 12/4.
4. 12/4 is 3 wholes, because 4/4 + 4/4 + 4/4 = 12/4. So 16 + 3 = 19.
5. The perimeter is 19 inches.
6. Check: four sides have 6 + 2 + 6 + 2 = 16 wholes and four 3/4, which is 12/4, or 3. 16 + 3 = 19.

Accept 16 12/4, 18 4/4 or 76/4.

Say it twice: length plus width is halfway around, so add it to itself.

12 min. Fix it: five wrong answers

Side 2 (Lia says: "A hall floor is 60 feet long and 9 feet wide. Its area is 138 square feet." **[60 × 9 = 540; 540 square feet. Lia found the edge (error A)]**, Ben says: "A track is 274 feet long and 189 feet wide. Its perimeter is 463 feet." **[274 + 189 = 463; 463 + 463 = 926; 926 feet. Ben added one length and one width (error B)]**, Kai says: "A patio covers 72 square feet. It is 8 feet wide, so it is 64 feet long." **[8 × ? = 72; 9 feet. Kai subtracted (error C)]**, Rosa says: "A goat pen has 58 feet of fence all the way around. It is 17 feet long, so it is 24 feet wide." **[17 + ? + 17 + ? = 58; 12 feet. Rosa did not halve (error D)]**, Nia says: "A stage floor is 50 feet long and 3 feet wide. Its area is 150 feet." **[50 × 3 = 150; 150 square feet. Nia named a length unit (error E)]**), one for each error A to E. Pairs, about 2 minutes each: sketch, say what went wrong, write the right working. Then the display shows the wrong working beside the right one.

Lia found the edge for a floor (A): ask "does the floor cover, or go around?" Ben stopped halfway around (B). Kai subtracted from an area (C): "8 times what is 72?" Rosa did not halve (D). Nia's number is right and her

unit is wrong (E): "is 150 feet a line or 150 squares?"

22 min. The generator sheet

Alone for 18 minutes, then 4 minutes checking in pairs. The sheet (sheet-d091) holds twelve mixed problems with no pictures. Every answer is a sketch with labelled sides, an equation and a unit. Most students finish 8 to 10 in 18 minutes. That is fine: the pairs check what is done, and the rest go home or into tomorrow's spare minutes.

At 18 minutes, partners swap and compare; where they disagree, both redo it. Then put the sheet's key on the display, equations and all. Students mark each answer with a tick, or a question mark and the right working.

Watch for: half-size perimeters (B); swapped measures (A); no unit, or feet for an area (E); a side from an area by taking away (C); a fence side not halved (D); a lost regrouped ten; denominators added. Ask: "Does this cover or go around? So which unit?"

If many make B: write 274 + 189 + 274 + 189 on the board beside 463 + 463 and show they match. If many make E: the eight answers with no units from the unit opener, each written as feet or square feet.

4 min. Pull it together

Whole class.

Ask. "When the sides are big, why add length plus width first?" Expect: "fewer numbers to add"; "the other half is the same". Show the first, then the second, then connect: 2 × (l + w) means the sum twice.

Ask. "With no picture, how do you know whether to write feet or square feet?" Expect: area covers, so it counts squares; perimeter goes around, so it is a length.

7 min. Exit: the slip

Alone, three questions (A hall is 40 feet long and 6 feet wide. Area? **[240 square feet]**, A park is 235 yards long and 148 yards wide. Perimeter? **[766 yards]**, A card is 3 1/10 inches long and 2 4/10 inches wide. Perimeter? **[11 inches; 10 10/10 is also right]**); students may sketch. On the record sheet under E91: a tick if all three are right; otherwise the letter of each error, and a dot for a number slip.

Other rooms

- **The sheet not printed.** Write the six problems from the sheet page's "Without a printer" list on the board; 12 minutes alone, then the key on the board. The extra time goes to the challenge.
- **One projector.** The display page runs the examples and the fixes; the class writes before each step shows.
- **One child at home.** Measure a door and a table top to the nearest foot with a tape measure or a ruler; find both measures of each, with units. Then do the sheet together.

Shorter or longer

Shorter: drop the pairs' check on the sheet (students check their own against the key) and the second Ask. Never drop the fixes, the sketches or the units, and never hand back the grid. Longer: the challenge on the board (A rectangle has an area of 60 square feet and a perimeter of 32 feet. How long are its sides? **[6 feet**

and 10 feet]), then the extension (Find every rectangle with whole-number sides and an area of 60 square feet. Find the perimeter of each. **[1 by 60, perimeter 122; 2 by 30, 64; 3 by 20, 46; 4 by 15, 38; 5 by 12, 34; 6 by 10, 32]**).

Words and supports

Word Wall cards: perimeter, area, square foot, mixed number, length, width. Frames on the display: "Length plus width is __, so the perimeter is __ + __ = __." "It covers, so the unit is square __." "It goes around, so the unit is __." A student who needs touch traces the four sides of the sketch with a finger while saying each number.

Tomorrow

On the slip, 92 for the hall (A): if many, before day 92's task, two minutes on Lia's fix. 383 or 5 5/10 (B): day 92's warm-up is two more big perimeters; if many, first show 463 + 463 beside 274 + 189 + 274 + 189. "240 feet" (E): if many, two minutes on the opener's eight answers with no units before day 92's warm-up. 24 or 2,400, 666 or 746, or a fraction slip in 11: practice, not a reteach. 10 10/10 is right; ask for it as 11.

Credit

Ours. Informed by, not adapted from: Illustrative Mathematics K-5, grade 4, unit 5 (CC BY 4.0, https://curriculum.illustrativemathematics.org/k5/), and the Progressions for the Common Core State Standards, Geometric Measurement.

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Day 91: Both formulas, big sides and fraction sides

Name _____ Date _____

I can use both formulas with no picture, with big sides and fraction sides.

Warm-up. Add.

1. 386 + 386 = ? _____	**2. 419 + 275 = ?** _____
3. $2\frac{3}{5} + 2\frac{3}{5} = ?$ _____	4. $3\frac{5}{8} + 1\frac{7}{8} = ?$ _____

Example 1. A school field is 264 feet long and 158 feet wide. How much fence goes all the way around?

Length + width: ____ + ____ = ____

Perimeter: ____ + ____ = ____ feet Check: _____

Example 2. A picture frame is $6\frac{3}{4}$ inches long and $2\frac{3}{4}$ inches wide. Find its perimeter.

Length + width: ____ + ____ = ____

Perimeter: ____ + ____ = ____ = ____ inches

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Fix it. Each answer is wrong. Sketch it. Say what went wrong. Then do it right.

1. Lia says: "A hall floor is 60 feet long and 9 feet wide. Its area is 138 square feet."

   What went wrong: _____ Right: _____

   Right: _____

2. Ben says: "A track is 274 feet long and 189 feet wide. Its perimeter is 463 feet."

   What went wrong: _____ Right: _____

   Right: _____

3. Kai says: "A patio covers 72 square feet. It is 8 feet wide, so it is 64 feet long."

   What went wrong: _____ Right: _____

   Right: _____

4. Rosa says: "A goat pen has 58 feet of fence all the way around. It is 17 feet long, so it is 24 feet wide."

   What went wrong: _____ Right: _____

   Right: _____

5. Nia says: "A stage floor is 50 feet long and 3 feet wide. Its area is 150 feet."

   What went wrong: _____ Right: _____

   Right: _____

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Day 91: Both formulas, big sides and fraction sides: key

Each answer is worked out in full. The last column names the wrong answers to expect and the error each one shows.

Where	Id	Part	Question	Answer	Wrong answers and what they mean
example	s11-K91-1	4.MD.A.3 (c), MP.6	A school field is 264 feet long and 158 feet wide. How much fence goes all the way around?	264 + 158 = 422; 422 + 422 = 844; 844 feet	422: error B, adds one length and one width (length plus width only)
example	s11-K91-2	4.MD.A.3 (d), MP.6	A picture frame is 6 3/4 inches long and 2 3/4 inches wide. Find its perimeter.	6 3/4 + 2 3/4 = 8 6/4; 8 6/4 + 8 6/4 = 16 12/4 = 19; 19 inches (16 12/4, 18 4/4 and 76/4 are also right)	9 2/4: error B, adds one length and one width (length plus width only)
fix	s11-F91-1	4.MD.A.3 (e), MP.6	Lia says: "A hall floor is 60 feet long and 9 feet wide. Its area is 138 square feet."	60 × 9 = 540; 540 square feet. Lia found the edge (error A)	138: error A, uses one formula for the other measure (the wrong answer shown)
fix	s11-F91-2	4.MD.A.3 (c), MP.6	Ben says: "A track is 274 feet long and 189 feet wide. Its perimeter is 463 feet."	274 + 189 = 463; 463 + 463 = 926; 926 feet. Ben added one length and one width (error B)	463: error B, adds one length and one width (the wrong answer shown)
fix	s11-F91-3	4.MD.A.3 (a), MP.2	Kai says: "A patio covers 72 square feet. It is 8 feet wide, so it is 64 feet long."	8 × ? = 72; 9 feet. Kai subtracted (error C)	64: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given (the wrong answer shown)
fix	s11-F91-4	4.MD.A.3 (b), MP.2	Rosa says: "A goat pen has 58 feet of fence all the way around. It is 17 feet long, so it is 24 feet wide."	17 + ? + 17 + ? = 58; 12 feet. Rosa did not halve (error D)	24: error D, from a perimeter, takes off one side, or does not halve (the wrong answer shown)
fix	s11-F91-5	4.MD.A.3 (g), MP.6	Nia says: "A stage floor is 50 feet long and 3 feet wide. Its area is 150 feet."	50 × 3 = 150; 150 square feet. Nia named a length unit (error E)	"150 feet": error E, wrong unit: "40 feet" for an area (the wrong answer shown)

practice	s11-P91-1	4.MD.A.3 (c)	386 + 386 = ?	772	
practice	s11-P91-2	4.MD.A.3 (c)	419 + 275 = ?	694	
practice	s11-P91-3	4.MD.A.3 (d)	2 3/5 + 2 3/5 = ?	5 1/5; 4 6/5 is also right	4 6/10: (denominators added: a fraction slip)
practice	s11-P91-4	4.MD.A.3 (d)	3 5/8 + 1 7/8 = ?	5 4/8; 4 12/8 and 5 1/2 are also right	
challenge	s11-C91-1	4.MD.A.3 (f), MP.7	A rectangle has an area of 60 square feet and a perimeter of 32 feet. How long are its sides?	6 feet and 10 feet	
challenge	s11-C91-2	4.MD.A.3 (g), MP.7	Find every rectangle with whole-number sides and an area of 60 square feet. Find the perimeter of each.	1 by 60, perimeter 122; 2 by 30, 64; 3 by 20, 46; 4 by 15, 38; 5 by 12, 34; 6 by 10, 32	
exit (paper)	s11-X91a	4.MD.A.3 (g)	A hall is 40 feet long and 6 feet wide. Area?	240 square feet	24 or 2,400: (a zero slip); 92: error A, uses one formula for the other measure; "240 feet": error E, wrong unit: "40 feet" for an area
exit (paper)	s11-X91b	4.MD.A.3 (c)	A park is 235 yards long and 148 yards wide. Perimeter?	766 yards	383: error B, adds one length and one width; 666 or 746: (a lost regrouped ten)
exit (paper)	s11-X91c	4.MD.A.3 (d)	A card is 3 1/10 inches long and 2 4/10 inches wide. Perimeter?	11 inches; 10 10/10 is also right	5 5/10: error B, adds one length and one width

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Name _____ Day 91

1. A hall is 40 feet long and 6 feet wide. Area?

_____

_____

2. A park is 235 yards long and 148 yards wide. Perimeter?

_____

_____

3. A card is $3\frac{1}{10}$ inches long and $2\frac{4}{10}$ inches wide. Perimeter?

_____

_____

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Name _____ Day 91

1. A hall is 40 feet long and 6 feet wide. Area?

_____

_____

2. A park is 235 yards long and 148 yards wide. Perimeter?

_____

_____

3. A card is $3\frac{1}{10}$ inches long and $2\frac{4}{10}$ inches wide. Perimeter?

_____

_____

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Name _____ Day 91

1. A hall is 40 feet long and 6 feet wide. Area?

_____

_____

2. A park is 235 yards long and 148 yards wide. Perimeter?

_____

_____

3. A card is $3\frac{1}{10}$ inches long and $2\frac{4}{10}$ inches wide. Perimeter?

_____

_____

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Name _____ Day 91

1. A hall is 40 feet long and 6 feet wide. Area?

_____

_____

2. A park is 235 yards long and 148 yards wide. Perimeter?

_____

_____

3. A card is $3\frac{1}{10}$ inches long and $2\frac{4}{10}$ inches wide. Perimeter?

_____

_____

Curio Class, grade 4 math, day 91, version draft 1, not yet taught, CC BY 4.0, curioclass.org

Day 91: the generator sheet

Make and print it: https://curioclass.org/closet/worksheets/make/measure-shapes/?grade=4&skill=formulas&count=12&fixes=off. Each press of New sheet gives a fresh sheet and its key; print one sheet per student and the key for you.

What it holds

Twelve problems in words, with no drawings, and no fixes: day 91 runs its own five fixes first. The twelve mix both formulas: an area from two sides, two areas with a side that is a multiple of 10, a missing side from an area, a missing side from a perimeter, a perimeter with two- and three-digit sides, and a perimeter with fraction sides (like denominators). The order is shuffled, so the words have to say which measure is asked for. The sheet says 18 minutes on its head.

The generator never uses a fact that the unit's lessons, exits or checks use: two sides with their area or their perimeter, in any order. So the sheet cannot hand students an answer they will meet on the end check. Each story thing comes in units and sizes it really has.

Making it

Open the link above, press New sheet once, and print one copy per student and the key for yourself. Every student gets the same sheet, so pairs can compare. The sheet code printed in its foot (five letters) reprints exactly the same sheet; write it on the day 91 teacher page if you want it again next year.

Using it on day 91

Alone for 18 minutes, then 4 minutes in pairs checking against each other; where partners disagree, both redo the problem. Then put the key up. Most students finish 8 to 10; the pairs check what is done.

Without a printer or a device

Write these six on the board; students work them in their notebooks. None uses a fact from the unit's lessons or checks (they are rows in the plan, so the gate checks them). The answers are in brackets.

1. A patio is 11 yards long and 6 yards wide. How much ground does it cover? **[11 × 6 = 66; 66 square yards]**
2. A poster covers 84 square inches. It is 7 inches wide. How long is it? **[7 × ? = 84; 12 inches]**
3. A goat pen uses 46 feet of fence. It is 15 feet long. How wide is it? **[15 + ? + 15 + ? = 46; 8 feet]**
4. A field is 236 meters long and 118 meters wide. How far is it around the edge? **[236 + 118 = 354; 354 + 354 = 708; 708 meters]**
5. A card is 4 1/8 inches long and 2 5/8 inches wide. Find its perimeter. **[13 4/8 inches; 12 12/8 is also right]**
6. A garden is 40 yards long and 8 yards wide. How much ground does it cover? **[40 × 8 = 320; 320 square yards]**

Curio Class, grade 4 math, unit 11.1, sheet-d091, version draft 1, not yet taught, CC BY 4.0, curioclass.org

Day 92: Area or perimeter?

Teacher page

Day	92, unit 11.1, use day
Goal	I can decide which measure a real problem needs
Standard parts	4.MD.A.3 (c) perimeter with multi-digit sides; 4.MD.A.3 (d) perimeter with fraction sides; 4.MD.A.3 (e) choosing the measure in a real problem; 4.MD.A.3 (f) the perimeter formula as 2 × (l + w): length plus width is half the perimeter
Tool	None
Materials	per student, grid paper (1 sheet) and a pencil; per class, the 36-foot string loop, the projector with the display page, and a board (no document camera: draw it on the board)
Print	student page, 1 per student, both sides; exit slip, 1 per student; the key, 1 for the teacher
Leans on	courses/math-4/u11-1/opener.md; courses/math-4/u11-1/record.md
Version	draft 1, not yet taught
Credit	ours

Before class

Print what the header lists. Have the 36-foot string loop and grid paper out, the display on the projector, and the two warm-up problems on the board.

The day

5 min. Warm-up: two perimeters

Alone, on the back of the grid paper. A rectangle is 264 feet long and 129 feet wide. Find its perimeter. **[786 feet]** and A rectangle is 2 3/5 inches long and 1 4/5 inches wide. Find its perimeter. **[8 4/5 inches; 6 14/5, 7 9/5 and 44/5 are also right]**. Students add the length and the width, then add the sum to itself, as on day 91. Go over both at 5 minutes. A sum not doubled (393, 4 2/5 or 3 7/5) is error B: trace the four sides on the board.

If many made B on day 91's exit (383 or 5 5/10): Ben's wrong answer from day 91's display first, one minute. **E** (240 feet): the opener's eight answers with no units replace A rectangle is 2 3/5 inches long and 1 4/5 inches wide. Find its perimeter. **[8 4/5 inches; 6 14/5, 7 9/5 and 44/5 are also right]**. **A** (92 for the hall): open the dog pen with Lia's wrong answer: "Does the floor cover, or go around?"

20 min. The dog pen, then teaching first

10 minutes: the task, in pairs (A dog pen is a rectangle. It uses all 36 feet of fence. Which pen gives the dog the most room? **[9 feet by 9 feet: 81 square feet. Every pen's length plus width is 18]**). Read the task aloud once. Four volunteers hold the corners of the string loop as a long thin pen, then a square one: same string, which dog has more room? Pairs sketch two pens on the grid side, the long thin one and

the square, and fill the rest of the table on side 1 by numbers alone: length + width must be 18. Watch for pairs who change the fence while changing the pen (a 10 by 10 pen): count its sides with them.

Ask. "Every pen used 36 feet of fence. What is the same about the length plus the width of each? Why?" Expect: "they all make 18"; then "the fence is two lengths and two widths, so one of each is half". Take the first, then press for the second.

Ask. "Two rectangles have the same perimeter. Must they have the same area?" Expect "no": point at 1 by 17 (17 square feet) and 9 by 9 (81).

Ask first, from the pairs' tables. Then close it, smallest pen first:

1. Every pen uses all 36 feet: two lengths and two widths.
2. So one length and one width use half the fence: 18 feet.
3. Every pen has length + width = 18: 1 and 17, 2 and 16, and so on up to 9 and 9.
4. Room for the dog is the inside: the area, length times width.
5. 1 × 17 = 17, 5 × 13 = 65, 8 × 10 = 80, 9 × 9 = 81.
6. The 9 by 9 pen gives the most room: 81 square feet.

10 minutes: two worked examples, whole class. Students fill the frames on side 1 as you go.

Example 1, times as long (A rug is 7 feet wide. Its length is 2 times its width. How much floor is under the rug? **[Length 2 × 7 = 14 feet; area 14 × 7 = 98 square feet]**).

1. Sketch the rug: 7 feet wide, the length unknown.
2. Draw the width as 1 bar of 7. The length is 2 bars of 7.
3. "2 times its width" means 2 × 7, not 7 + 2. The rug is 14 feet long.
4. The floor under the rug is the inside of the rectangle: the area.
5. Area is length times width: 14 × 7 = 98.
6. The rug is on 98 square feet of floor. Square feet, because it is area.
7. Check: 14 is 2 sevens, and 98 is 14 rows of 7.

"2 times as long" is 2 × 7, not 7 + 2: the same as stretch 10's comparing. A child who writes 9 is adding.

Example 2, two steps (A chicken yard has 72 square feet of ground. It is 8 feet wide. How much fence does it need? **[8 × ? = 72, so 9 feet long; fence 9 + 8 + 9 + 8 = 34 feet]**).

1. Sketch the yard: 8 feet wide, the length unknown, 72 square feet inside.
2. Step 1: the 72 square feet is the inside, so 8 × ? = 72. 8 × 9 = 72: the yard is 9 feet long.
3. Step 2: fence goes along the edge of the yard. That is the perimeter.
4. 9 + 8 + 9 + 8 = 34, or 2 × (9 + 8) = 34.
5. The yard needs 34 feet of fence. Feet, because it is a length.

In each step ask: "Is this number the inside or the edge?"

20 min. Six problems by hand

Alone, side 2. Read all six aloud once at the start. For each, students write "area" or "perimeter" first, sketch on the back of the grid paper (a plain rectangle with its sides labelled; do not count squares), then the answer with its unit. None of the six says "covers" or "around": the story tells which. A kitchen floor is 9 feet

long and 6 feet wide. Each tile is 1 foot by 1 foot. How many tiles does the floor need? **[9 × 6 = 54 tiles (area, 54 square feet)]** Zoe makes a rectangle out of tape on the gym floor. It is 14 feet long and 9 feet wide. How much tape does she use? **[14 + 9 + 14 + 9 = 46 feet (perimeter)]** A bench seat is 2 feet wide. It is 6 times as long as it is wide. Mom paints the top of the seat. How much does she paint? **[Length 6 × 2 = 12 feet; 12 × 2 = 24 square feet (area)]** A dog yard is 6 feet wide. It has 42 square feet of ground. How much fence closes it in? **[6 × ? = 42, so 7 feet long; fence 7 + 6 + 7 + 6 = 26 feet]** A flag is 9 inches wide. It is 2 times as long as it is wide. Gold trim makes a border on the flag. How much trim is used? **[Length 2 × 9 = 18 inches; 18 + 9 + 18 + 9 = 54 inches (perimeter)]** Kai bends 24 inches of wire into a rectangle. It is 7 inches long. How wide is it? **[7 + ? + 7 + ? = 24; 5 inches (the wire is the perimeter)]**

After 15 minutes, pairs swap and check each answer against the story.

Check: Is your number the inside or the edge? Is the unit right? If not, put a question mark.

Watch for: a "times as long" length found by adding (8 for the bench, 11 for the flag); the wrong measure (30 tiles, 126 feet of tape, 162 inches of trim: error A); "54 feet" of tiles (error E); stopping at the side (7 for the dog yard). If many choose the wrong measure: each pair sketches problem 1 and shades what the story asks for, the inside or the edge. If many add for "times as long": draw 2 bars of 9 for the flag.

5 min. Written task

Alone, the box at the foot of side 2: Look at problem 2. Did it need area or perimeter? How did you know? **[Perimeter, 46 feet: the tape makes the sides, not the floor inside]**. Students may answer aloud to you instead (a scribe is allowed). Collect the student pages with the exit slips at the end; read after class.

Tick each look-for on the key alone; the key gives samples that meet, partly meet and do not (ours, untested). A child who writes "tape, lines, 46 feet" with no full sentence has look-fors 1 to 3: that is language, not math.

5 min. Pull it together

Whole class.

Ask. "Which problem was hardest to decide? What in the story told you?" Expect the dog yard or the flag. Show one decided from the story, then one from a word; connect them: "inside or edge?" before any number.

5 min. Exit: the slip

Alone, on paper: A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units. **[48 square yards; 32 yards]**. Read it aloud once. Write the results on the record sheet under E92.

Other rooms

- **No string loop.** Draw a 17 by 1 pen and a 9 by 9 pen on the board, each with 36 feet of edge.
- **One child at home.** Make the 36-foot pens from a 36-inch string on a table, with paper squares for the room inside. Then do problems 1 to 3 together and 4 to 6 alone.

Shorter or longer

Shorter: stop the task's table at 6 pens and close it from there; drop the partner check (students check their own). Never drop the written task or the reading aloud. Longer: the challenge (A dog pen is 10 feet long and 2 feet wide. Draw a pen with more room but less fence. **[For example 5 by 5: 25 square feet and 20 feet of fence. The 10 by 2 pen has 20 square feet and 24 feet of fence]**), on the back of the grid paper.

Words and supports

Word Wall cards: area, perimeter, length, width, square foot, times as long. Frames on the display: "This problem needs __ because the story asks about the __." "The length is __ times __, so it is __ feet."

Tomorrow

Day 93 is the end check; the check sets its groups. Mark E92. A length of 7 (28 and 22) is adding for "times as long": those students join day 94's first five minutes with end item 5's "length of 9" group (3 bars of 4). Swapped measures (32 for the floor, 48 for the fence: A) or "48 yards" (E): a student who also misses end item 7 is in its group already. Hand back the written tasks at the door, look-fors ticked; missing look-for 1 or 4 joins the item 7 group in the respond half.

Credit

Ours. Informed by, not adapted from: Illustrative Mathematics K-5, grade 4, unit 5 (CC BY 4.0, https://curriculum.illustrativemathematics.org/k5/), and the Progressions for the Common Core State Standards, Geometric Measurement.

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Day 92: Area or perimeter?

Name _____ Date _____

I can decide which measure a real problem needs.

The dog pen. A dog pen is a rectangle. It uses all 36 feet of fence. Which pen gives the dog the most room?

Length									
Width									
Length + width									
Room inside									

The most room: ___ feet by ___ feet, ___ square feet.

Example 1. A rug is 7 feet wide. Its length is 2 times its width. How much floor is under the rug?

Length: ___ × ___ = ___ Area or perimeter? _____ Answer: ___ _____

Example 2. A chicken yard has 72 square feet of ground. It is 8 feet wide. How much fence does it need?

Step 1: ___ × ___ = ___ Step 2: ___ + ___ + ___ + ___ = ___ Answer: ___ _____

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Six problems. Write area or perimeter first. Then solve. Write the unit.

1. A kitchen floor is 9 feet long and 6 feet wide. Each tile is 1 foot by 1 foot. How many tiles does the floor need?

   Area or perimeter? _____ Answer: _____

2. Zoe makes a rectangle out of tape on the gym floor. It is 14 feet long and 9 feet wide. How much tape does she use?

   Area or perimeter? _____ Answer: _____

3. A bench seat is 2 feet wide. It is 6 times as long as it is wide. Mom paints the top of the seat. How much does she paint?

   Area or perimeter? _____ Answer: _____

4. A dog yard is 6 feet wide. It has 42 square feet of ground. How much fence closes it in?

   Area or perimeter? _____ Answer: _____

5. A flag is 9 inches wide. It is 2 times as long as it is wide. Gold trim makes a border on the flag. How much trim is used?

   Area or perimeter? _____ Answer: _____

6. Kai bends 24 inches of wire into a rectangle. It is 7 inches long. How wide is it?

   Area or perimeter? _____ Answer: _____

Tell me. Look at problem 2. Did it need area or perimeter? How did you know?

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Day 92: Area or perimeter?: key

Each answer is worked out in full. The last column names the wrong answers to expect and the error each one shows.

Where	Id	Part	Question	Answer	Wrong answers and what they mean
example	s11-K92-1	4.MD.A.3 (f), MP.4	A dog pen is a rectangle. It uses all 36 feet of fence. Which pen gives the dog the most room?	9 feet by 9 feet: 81 square feet. Every pen's length plus width is 18	every pen has the same room: (same fence taken as same area)
example	s11-K92-2	4.MD.A.3 (e), MP.4	A rug is 7 feet wide. Its length is 2 times its width. How much floor is under the rug?	Length 2 × 7 = 14 feet; area 14 × 7 = 98 square feet	length 9, area 63: ("2 more", stretch 10); 42: error A, uses one formula for the other measure
example	s11-K92-3	4.MD.A.3 (e), MP.4	A chicken yard has 72 square feet of ground. It is 8 feet wide. How much fence does it need?	8 × ? = 72, so 9 feet long; fence 9 + 8 + 9 + 8 = 34 feet	9: (stops at the side); 17: error B, adds one length and one width
practice	s11-P92-1	4.MD.A.3 (c), MP.6	A rectangle is 264 feet long and 129 feet wide. Find its perimeter.	786 feet	393: error B, adds one length and one width (Adding slips go to the warm-up)
practice	s11-P92-2	4.MD.A.3 (d), MP.6	A rectangle is 2 3/5 inches long and 1 4/5 inches wide. Find its perimeter.	8 4/5 inches; 6 14/5, 7 9/5 and 44/5 are also right	4 2/5: error B, adds one length and one width; 3 7/5: error B
practice	s11-P92-3	4.MD.A.3 (e), MP.4	A kitchen floor is 9 feet long and 6 feet wide. Each tile is 1 foot by 1 foot. How many tiles does the floor need?	9 × 6 = 54 tiles (area, 54 square feet)	30: error A, uses one formula for the other measure; "54 feet": error E, wrong unit: "40 feet" for an area
practice	s11-P92-4	4.MD.A.3 (e), MP.4	Zoe makes a rectangle out of tape on the gym floor. It is 14 feet long and 9 feet wide. How much tape does she use?	14 + 9 + 14 + 9 = 46 feet (perimeter)	126: error A, uses one formula for the other measure; 23: error B, adds one length and one width
practice	s11-P92-	4.MD.A.3 (e), MP.4	A bench seat is 2 feet wide. It is 6 times as	Length 6 × 2 = 12 feet; 12 × 2 = 24 square feet	length 8, area 16: ("6 more", stretch 10);

	5		long as it is wide. Mom paints the top of the seat. How much does she paint?	(area)	28: error A, uses one formula for the other measure
practice	s11-P92-6	4.MD.A.3 (e), MP.4	A dog yard is 6 feet wide. It has 42 square feet of ground. How much fence closes it in?	6 × ? = 42, so 7 feet long; fence 7 + 6 + 7 + 6 = 26 feet	7: (stops at the side); 96: error A, uses one formula for the other measure (the area used as a side)
practice	s11-P92-7	4.MD.A.3 (e), MP.4	A flag is 9 inches wide. It is 2 times as long as it is wide. Gold trim makes a border on the flag. How much trim is used?	Length 2 × 9 = 18 inches; 18 + 9 + 18 + 9 = 54 inches (perimeter)	length 11, trim 40: ("2 more", stretch 10); 162: error A, uses one formula for the other measure
practice	s11-P92-8	4.MD.A.3 (e), MP.4	Kai bends 24 inches of wire into a rectangle. It is 7 inches long. How wide is it?	7 + ? + 7 + ? = 24; 5 inches (the wire is the perimeter)	17: error D, from a perimeter, takes off one side, or does not halve (one long side taken off); 10: error D (not halved)
practice	s11-P92-9	4.MD.A.3 (e), MP.3	Look at problem 2. Did it need area or perimeter? How did you know?	Perimeter, 46 feet: the tape makes the sides, not the floor inside **Look-fors, each ticked alone:** 1. Says what is measured: the inside or the edge. 2. Names the measure. 3. The unit matches: feet for perimeter. 4. The reason comes from the situation, not from one word. **Samples (ours, untested):** • meets: Perimeter. The tape makes the lines of the rectangle. It does	

				not fill the floor. 14 + 9 + 14 + 9 = 46 feet. • partly: Perimeter, because it is tape. (Look-fors 2 and 3 only: ask what the tape does.) • not: Area, because it says floor. 14 × 9 = 126 square feet. (It rests on one word and measures the inside.)	
challenge	s11-C92-1	4.MD.A.3 (f), MP.3	A dog pen is 10 feet long and 2 feet wide. Draw a pen with more room but less fence.	For example 5 by 5: 25 square feet and 20 feet of fence. The 10 by 2 pen has 20 square feet and 24 feet of fence	
exit (paper)	s11-X92	4.MD.A.3	A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units.	48 square yards; 32 yards	length 7: ("3 more", stretch 10); swapped: error A, uses one formula for the other measure; "48 yards": error E, wrong unit: "40 feet" for an area

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Name _____ Day 92

1. A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units.

_____

_____

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Name _____ Day 92

1. A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units.

_____

_____

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Name _____ Day 92

1. A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units.

_____

_____

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Name _____ Day 92

1. A patio is 4 yards wide. Its length is 3 times its width. (a) Ana paints the patio. How much does she paint? (b) She fences in the patio. How much fence does she need? Write the units.

_____

_____

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Day 93: Show what I know

Teacher page

Day	93, unit 11.1, check day
Goal	I can show what I know and fix what I missed
Standard parts	4.MD.A.3 (a) the area formula, a side unknown; 4.MD.A.3 (b) the perimeter formula, a side unknown; 4.MD.A.3 (c) perimeter with multi-digit sides; 4.MD.A.3 (d) perimeter with fraction sides; 4.MD.A.3 (e) choosing the measure in a real problem; 4.MD.A.3 (f) the perimeter formula as 2 × (l + w): length plus width is half the perimeter; 4.MD.A.3 (g) the area formula, both sides known
Tool	None
Materials	per student, a pencil and a pen or coloured pencil for marking; per class, the projector with the display page (or the board), grid paper for the groups
Print	student page (the end check), 1 per student, both sides; large print on request; the key, 1 for the teacher; the record sheet, 1
Leans on	courses/math-4/u11-1/opener.md; courses/math-4/u11-1/record.md
Uses when it exists	courses/math-4/u12-1/readiness.md
Version	draft 2, not yet taught
Credit	ours

Before class

Print the end check both sides, the key and the record sheet. Open the display; show the answers only at the marking. Put grid paper out for the groups.

The day

30 min. End check

Alone, unaided: no grid paper, tiles or times table. Read each item aloud once. Early finishers check each unit. At 30 minutes, pencils down. An item not reached is recorded as not reached, not wrong.

20 min. Respond: mark, then fix in groups

Mark, 5 minutes. Pencils away; show the answers to 1 to 4 and 7. Students tick or circle each in pen or coloured pencil. Items 5 and 6 come back in the last ten minutes.

Groups, 15 minutes. Hands up by item sets the groups. A student who missed several joins the group for the first one missed. Each group fixes its item in pen beside the first try, then checks it: put the side back in the equation, or add the four sides again. Each group's task is on the display.

- Item 1 (error C): rows of 4 on grid paper until the 28 squares are used.

- Item 2 (error D): fill in 16 + __ + 16 + __ = 50 together.
- Items 3 and 4 (error B): trace and name all four sides, then write l + w + l + w. A right setup with a wrong sum is an adding slip: redo the sum only.
- Item 7 (error A or E): sketch it, shade the inside, trace the edge in another colour. Area is the shaded part, in square centimeters. A student whose day 92 written task missed look-for 1 or 4 joins this group.

Students who missed nothing do the challenge (A rectangle has an area of 90 square feet and a perimeter of 38 feet. How long are its sides? **[9 feet and 10 feet]**), then the extension (Find every rectangle with whole-number sides and an area of 42 square feet. Find the perimeter of each. **[1 by 42, 86; 2 by 21, 46; 3 by 14, 34; 6 by 7, 26]**), then check a group's fixes. Collect the checks.

10 min. Items 5 and 6 together

Hand back the checks. Pairs read each other's items 5 and 6 and ask the two questions on the display. Students add to their own answers in pen; you judge the pencil first try after class.

Ask. "Which part asked for the inside, and which for the edge?"

Ask. "Do your two sides add to 19? Why does that make the perimeter the same?"

When stretch 12's readiness check exists, it takes these ten minutes and items 5 and 6 return on day 94 instead.

Other rooms

- **No projector.** Write the answers and the group tasks on a covered part of the board.
- **One child at home.** Give the check, then go over each missed item with paper squares or string.

Shorter or longer

Shorter: cut the groups to ten minutes, never the check or the marking. Longer: the extension (Find every rectangle with whole-number sides and an area of 42 square feet. Find the perimeter of each. **[1 by 42, 86; 2 by 21, 46; 3 by 14, 34; 6 by 7, 26]**) for everyone who finishes a fix.

Words and supports

The defaults hold: items read aloud once, large print offered, a scribe or an oral answer on items 5 and 6, a sketch drawn from a student's words.

Tomorrow

About 40 minutes after class: record 1 to 4 and 7 from the pencil first try, then judge 5 and 6, ticking each look-for alone.

Item 5:

1. finds the length, 4 × 5 = 20
2. uses area for the sheet and perimeter for the border (part e)
3. writes square feet for the sheet and feet for the border

Item 6:

1. gives two sides that add to 19 and are not 11 and 8 (fractions count, and so does 9 1/2 by 9 1/2)
2. gives a reason that rests on length plus width, on half the perimeter, or on one side 1 longer and the other 1 shorter (part f)

A length of 9 ("4 more"): draw 4 bars of 5 beside one bar of 5. "20 by 5" for (a) is asked again: "How much plastic is that?"

Credit

Ours. Informed by, not adapted from: the Progressions for the Common Core State Standards, Geometric Measurement.

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Day 93: Show what I know

Name _____ Date _____

I can show what I know and fix what I missed.

Work alone. Show your work. Write the units.

1. A closet floor covers 28 square feet. It is 4 feet wide. How long is it? Write an equation with ? for the unknown side.

Equation: _____ Answer: _____

2. Mia uses all 50 feet of her fence to make a rectangular dog pen. The pen is 16 feet long. How wide is it? Show your steps.

Answer: _____

3. A rectangle is $2\frac{3}{8}$ inches long and $1\frac{1}{8}$ inches wide. Find its perimeter.

Answer: _____

4. A rectangle is 318 meters long and 147 meters wide. Find its perimeter.

Answer: _____

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5. A sandbox is 5 feet wide. It is 4 times as long as it is wide. (a) A plastic sheet lies flat on top and just fits. How much plastic is that? (b) A wooden border runs along all four sides. How long is the border? Write the units.

(a) _____ (b) _____

6. A garden is 11 feet long and 8 feet wide. Write the length and width of a different rectangle with the same perimeter. How can you tell the perimeters match without adding all four sides?

Length: _____ Width: _____

How I know: _____

_____

7. A rectangle is 70 centimeters long and 8 centimeters wide. Find its area.

Answer: _____

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Day 93: Show what I know: key

Each answer is worked out in full. The last column names the wrong answers to expect and the error each one shows.

Where	Id	Part	Question	Answer	Wrong answers and what they mean
end 1	s11-E1	4.MD.A.3 (a), MP.2	A closet floor covers 28 square feet. It is 4 feet wide. How long is it? Write an equation with ? for the unknown side.	4 × ? = 28 (also right: ? × 4 = 28, 28 = 4 × ?, 28 ÷ 4 = ?); 7 feet	24 or 32: error C, finds a side from an area by subtracting, adding, or multiplying the two numbers given; 10: error A, uses one formula for the other measure; 112: error C
end 2	s11-E2	4.MD.A.3 (b), MP.2	Mia uses all 50 feet of her fence to make a rectangular dog pen. The pen is 16 feet long. How wide is it? Show your steps.	9 feet	34 or 18: error D, from a perimeter, takes off one side, or does not halve; 17: error D (one side taken off, then halved)
end 3	s11-E3	4.MD.A.3 (d), MP.6	A rectangle is 2 3/8 inches long and 1 1/8 inches wide. Find its perimeter.	7 inches; 6 8/8 and 56/8 are also right	3 4/8: error B, adds one length and one width (Fraction slips go to the warm-up); 6 8/16: (added the denominators; fraction slips go to the warm-up)
end 4	s11-E4	4.MD.A.3 (c), MP.6	A rectangle is 318 meters long and 147 meters wide. Find its perimeter.	930 meters	465: error B, adds one length and one width (Adding slips go to the warm-up); 783: (one side doubled only (from memory))
end 5	s11-E5	4.MD.A.3 (e), MP.4	A sandbox is 5 feet wide. It is 4 times as long as it is wide. (a) A plastic sheet lies flat on top and just fits. How much plastic is that? (b) A wooden border runs along all four sides. How long is	100 square feet; 50 feet **Look-fors, each ticked alone:** 1. finds the length, 4 × 5 = 20 2. uses area for the sheet and perimeter for	Length 9, giving 45 and 28: ("4 more", stretch 10); swapped: error A, uses one formula for the other measure; 25: error B, adds one length and one width; "100 feet": error E, wrong unit: "40 feet" for an area

			the border? Write the units.	the border (part e) 3. writes square feet for the sheet and feet for the border **Samples (ours, untested):** • meets: Length 4 × 5 = 20. Sheet 5 × 20 = 100 square feet. Border 20 + 5 + 20 + 5 = 50 feet. • partly: 5 + 4 = 9. Sheet 45 square feet, border 28 feet. (Look-for 2 is met; the length goes to the comparing help: 4 bars of 5.) • not: Length 20. Sheet 20 + 5 + 20 + 5 = 50 feet. Border 20 × 5 = 100 feet.	
end 6	s11-E6	4.MD.A.3 (f), MP.3	A garden is 11 feet long and 8 feet wide. Write the length and width of a different rectangle with the same perimeter. How can you tell the perimeters match without adding all four sides?	For example 12 by 7; length plus width is still 19 **Look-fors, each ticked alone:** 1. gives two sides that add to 19 and are not 11 and 8 (fractions count, and so does 9 1/2 by 9 1/2) 2. gives a reason that rests on length plus	8 by 11: (the same rectangle); sides with the same area, such as 22 by 4; four sides added

				width, on half the perimeter, or on one side 1 longer and the other 1 shorter (part f) **Samples (ours, untested):** • meets: 12 by 7, because 11 + 8 = 19 and 12 + 7 = 19, and the perimeter is double that. • partly: 10 by 9. I added 10 + 9 + 10 + 9 and got 38. (Look-for 1 only.) • not: 22 by 4, because 22 × 4 = 88 and 11 × 8 = 88.	
end 7	s11-E7	4.MD.A.3 (g), MP.6	A rectangle is 70 centimeters long and 8 centimeters wide. Find its area.	560 square centimeters	156: error A, uses one formula for the other measure; 56 or 5,600: (a zero slip); "560 centimeters": error E, wrong unit: "40 feet" for an area
challenge	s11-C93-2	4.MD.A.3 (f), MP.2	A rectangle has an area of 90 square feet and a perimeter of 38 feet. How long are its sides?	9 feet and 10 feet	
challenge	s11-C93-3	4.MD.A.3 (g), MP.7	Find every rectangle with whole-number sides and an area of 42 square feet. Find the perimeter of each.	1 by 42, 86; 2 by 21, 46; 3 by 14, 34; 6 by 7, 26	

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Unit 11.1 record sheet: area and perimeter formulas

One row a student; print two for a class over 30. Each cell gets a tick; an error letter A to F (unit opener); s for a slip (a fact, adding, fraction or zero slip with the right setup); x for any other wrong answer; or a dash for not reached. R1 to R4: the readiness check, day 88. E89 to E92: the exits; on day 90 a tick means both parts, by either path. End 1 to 7: the end check, day 93; items 5 and 6 take one tick for each look-for on the key (listed under the table). Each head names the standard's part it records.

Student	R1	R2	R3	R4	E89	E90 (a, b)	E91 (c, d, g)	E92	End 1 (a)	End 2 (b)	End 3 (d)	End 4 (c)	End 5 (e)			End 6 (f)		End 7 (g)
													1	2	3	1	2	

End 5 look-fors: 1. finds the length, 4 × 5 = 20; 2. uses area for the sheet and perimeter for the border (part e); 3. writes square feet for the sheet and feet for the border

End 6 look-fors: 1. gives two sides that add to 19 and are not 11 and 8 (fractions count, and so does 9 1/2 by 9 1/2); 2. gives a reason that rests on length plus width, on half the perimeter, or on one side 1 longer and the other 1 shorter (part f)

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Unit 11.1 family note: area and perimeter

English

This week your child is learning two ways to measure a rectangle.

Area covers. It is how many squares fit inside: a rug 6 feet by 8 feet covers 6 × 8 = 48 square feet.

Perimeter goes around. It is the length of the edge: the same rug has 6 + 8 + 6 + 8 = 28 feet of edge.

Your child will also work backwards. If a rug covers 48 square feet and is 6 feet wide, how long is it? (8 feet, because 6 × 8 = 48.)

Try this at home, five minutes. Ask your child for both measures of your kitchen table, a book or a doormat. Measure the sides together with anything you have, even hand spans. Ask: "Which number tells how much it covers? Which tells how far it is around?" The answer to the first is in square units; the second is not.

No worksheet to return.

Spanish

Esta semana su hijo o hija aprende dos maneras de medir un rectángulo.

El área cubre. Es cuántos cuadrados caben adentro: una alfombra de 6 pies por 8 pies cubre 6 × 8 = 48 pies cuadrados.

El perímetro da la vuelta. Es el largo del borde: la misma alfombra tiene 6 + 8 + 6 + 8 = 28 pies de borde.

También trabajará al revés. Si una alfombra cubre 48 pies cuadrados y mide 6 pies de ancho, ¿cuánto mide de largo? (8 pies, porque 6 × 8 = 48.)

En casa, cinco minutos. Pídale las dos medidas de la mesa de la cocina, un libro o un tapete. Midan los lados juntos con lo que tengan, hasta con palmos. Pregunte: "¿Qué número dice cuánto cubre? ¿Cuál dice cuánto mide alrededor?" La respuesta a la primera va en unidades cuadradas; la segunda, no.

No hay hoja que devolver.

Curio Class, grade 4 math, unit 11.1, family, version draft 1, not yet sent; the Spanish needs a qualified reader before it goes home, CC BY 4.0, curioclass.org